Data-Driven PDF Method for Langevin Equations with Colored Noise

Hongli Zhao

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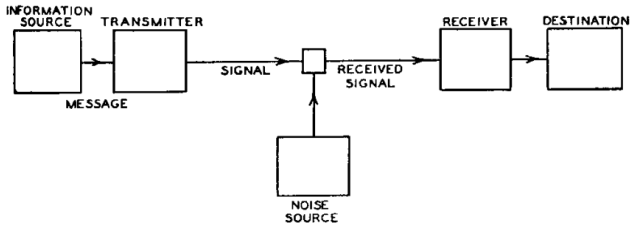
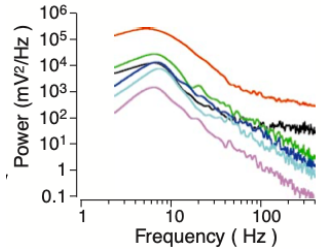
Topics

1. Uncertainty Quantification
2. PDF Method
3. Data-Driven PDF Discovery

Uncertainty Quantification (UQ)

- Mathematical models suffer from Langevin forces and parametric uncertainty
  - Motion of a free Brownian particle immersed in water
  - Noisy channel over a communication network
  - Random release of neurotransmitters by synapses in neurons





- Examples:
  (Langevin equations)

$$\frac{d}{dt}u = f(u,t) + g(u,t)\cdot\xi(t,\omega) \tag{1}$$

(advection-reaction PDE)

$$\frac{\partial}{\partial t}u + a(x,\omega)\cdot\frac{\partial}{\partial x}u = b(x,\omega)\cdot f(u) \tag{2}$$

with possibly random initial and boundary conditions.

- A model driven by random forces admits multiple solutions
- The method of distributions aims to derive a deterministic PDE for either the PDF or the CDF.

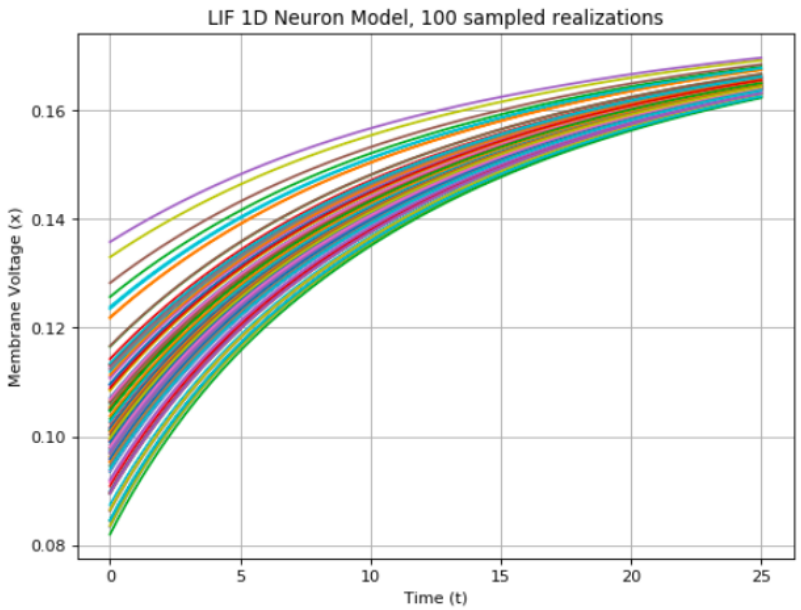


Figure: realizations of the Leaky Integrate-and-Fire model with random initial conditions, with IC drawn from a Gaussian distribution and computed using *RK4*

Method of Distributions

- Assume some additional constraints guaranteeing existence and uniqueness, the above cases admit *multiple solutions*.
- Instead of considering the paths directly, we can characterize the solutions with a PDF $p(u; x, t)$.
- We focused on Langevin equations, for arbitrary $\xi(t, \omega)$, using a Taylor series expansion, we have the *Kramers-Moyal expansion* for the probability density:

$$\frac{\partial}{\partial t} p(U; x, t) = \sum_{i=1}^{\infty} (-\frac{\partial}{\partial x})^i D^{(i)}(x, t) p(U; x, t) \tag{3}$$

  where $D^{(n)}(x, t) = \frac{1}{n!} \lim_{\Delta t \to 0} \frac{1}{\Delta t} M_n(x, t, \Delta t)$, M_n is the n-th moment.
- If ξ is Gaussian white noise, all higher order coefficients $D^{(n)}, n > 2$ go to 0, we recover the *Fokker-Planck equation*.

PDF Method

Problem

Consider the system of SODEs in $\mathbb{R}^N$:

$$\frac{dX(t,\omega)}{dt} = v(X(t,\omega), t; \xi(t,\omega))$$

$$X(0,\omega) = X^0(\omega)$$

where $v : \mathbb{R}^N \times \mathbb{R}^+ \mapsto \mathbb{R}$ is deterministic and parametrized by the random coefficient $\xi(t,\omega)$.

- We also assume v satisfies conditions to guarantee a unique pathwise solution.
  - J.L Strand, Random ordinary differential equations, Journal of Differential Equations
- Parameter class $\xi(t,\omega)$ is characterized by correlation time τ.

Derivation of PDF Method

Define a "raw PDF" which is a *Dirac Delta function* with respect to our solution to the SDE, X:

$$\Pi(x,t;\omega) = \delta(x - X(t,\omega)) = \prod_{i=1}^{N} \delta(x_i - X_i(t,\omega))$$

- By the sifting property, the ensemble mean over all realizations of our state variable x for the "raw PDF" gives the exact probability density at x.

$$\langle \Pi(x,t) \rangle = \int_{\mathbb{R}^N} \delta(y-x) p(y,t) dy = p(x,t)$$

- By construction, $\Pi(x,t;\omega)$ has the *Reynold's decomposition*:

$$\Pi = \langle \Pi \rangle + \Pi' = p + \Pi'$$

giving a zero mean stochastic fluctuation $\langle \Pi' \rangle = \langle \Pi - p \rangle = 0$.

- Weakly differentiate Π w.r.t. time t gives:

$$\frac{\partial \Pi}{\partial t} = -\frac{dX_i(t)}{dt} \cdot \delta'(x_i - X_i(t))$$

where δ' is the weak derivative of δ, recall $\delta'(-x) = -\delta'(x)$.

- Furthermore:

$$= -v_i(X_i(t), t)\frac{\partial \Pi}{\partial x_i} = -\nabla_x \cdot (v(x,t)\Pi(x,t))$$

using the sifting property of *dirac delta*.

- We arrived at the following conservation law:

$$\frac{\partial \Pi}{\partial t} + \nabla_x \cdot (v\Pi) = 0$$

- Using the Reynolds' decompositions, $v = \langle v \rangle + v', \Pi = \langle \Pi \rangle + \Pi'$, and $\langle \Pi \rangle = p$, both v', Π' have zero mean. Take ensemble mean on both sides and apply product rule, we arrive at the governing equation for our PDF:

$$\frac{\partial p}{\partial t} + \nabla_x \cdot (\langle v \rangle p) + \nabla_x \cdot \langle v'\Pi' \rangle = 0$$

The PDF Equation

IBVP

As above, we arrive at the following IBVP:

$$\frac{\partial p}{\partial t} + \nabla_x \cdot (\langle v \rangle p) + \nabla_x \cdot \langle v' \Pi' \rangle = 0 \tag{4}$$

with initial condition:

$$p(x, 0) = p_0(x)$$

which we know from the initial distribution for our SDE X_0. and vanishing free-space boundary condition as $x_i \to \pm\infty$.

- If v' is Gaussian white noise, the PDF method is exact and we recover the Fokker-Planck equation.
- In general, the cross variance term $\langle v' \Pi' \rangle$ is unknown. Computing $\langle v' \Pi' \rangle$ would require the knowledge of the solution X. The system is unclosed.

Closure approximation

- Stochastic problems are solved usually via Monte-Carlo simulations by:
  1. generating multiple realizations for the random parameters
  2. solving the appropriate deterministic PDE using these parameters
  3. computing ensembles for the multiple realizations
- However, MCS is known to have slow convergence; the computational cost also grows combinatorially w.r.t. the number of dimensions
  - Would like to devise more efficient numerical techniques
- One class of techniques is via deriving *analytical closure approximations* for the term $\langle v'\Pi' \rangle$.

LED closure approximation

A family of closure approximation is named the Large-Eddy-Diffusivity closure, originally motivated by the statistical study of turbulence flow problems.

- Hudong Chen, Shiyi Chen, and Robert H. Kraichnan, Phys. Rev. Lett. 63, 2657
- Let L denote the deterministic linear differential operator:

$$L(\Pi) = \frac{\partial \Pi}{\partial t} + \nabla_x \cdot (\langle v \rangle \Pi)$$

- Consider the terminal problem defined by the adjoint operator $\hat{L}$:

$$\langle L\Pi, G \rangle = \left\langle \Pi, \hat{L}G \right\rangle$$

i.e.

$$\hat{L}G = -\frac{\partial G}{\partial s} - \langle v \rangle \cdot \nabla_y G = \delta(x-y)\delta(t-s) \tag{5}$$

- G is the solution to the adjoint problem given by the Green's function with *dirac delta* source.

- We can solve the above problem using method of characteristics:

$$\frac{d\phi}{dr} = \langle v(\phi, r) \rangle$$

with initial data:

$$\phi(s) = y$$

where $\phi(r) \equiv \phi(r; y, s)$ is the flow map.

- Along the characteristic lines, we have an ODE for G:

$$\frac{dG(x, t; \phi(r, y, s), s)}{dr} = -\delta(t - r)\delta(x - \phi(r, y, s))$$

with terminal data:

$$G(x, t; \phi(t, y, s), t) = 0$$

- The solution for G along the characteristics is:

$$G(x, t; y, s) = \mathcal{H}(t - s)\delta(x - \phi(t; y, s))$$

- Apply the linear operator L on Π' which we wish to isolate, we perform some simple substitutions:

$$L\Pi' = L(\Pi - \langle \Pi \rangle) = L\Pi - L\langle \Pi \rangle$$

$$= \frac{\partial \Pi}{\partial t} + \nabla_x \cdot (\langle v \rangle \Pi) - (\frac{\partial \langle \Pi \rangle}{\partial t} + \nabla_x \cdot (\langle v \rangle \langle \Pi \rangle))$$

$$= \frac{\partial \Pi}{\partial t} + \nabla_x \cdot (\langle v \rangle \Pi) - (\frac{\partial p}{\partial t} + \nabla_x \cdot (\langle v \rangle p))$$

$$= -\nabla_x \cdot (v\Pi) + \nabla_x \cdot (\langle v \rangle \Pi) - (-\nabla_x \cdot (\langle v \rangle p) - \nabla_x \cdot \langle v'\Pi' \rangle + \nabla_x \cdot (\langle v \rangle p))$$

$$= -\nabla_x \cdot (v'\Pi - \langle v'\Pi' \rangle)$$

- Integrate by parts and use property of Green's function:

$$\int_0^t \int_{\mathbb{R}^N} GL\Pi' dyds = -\int_{\mathbb{R}^N} G(x,t;y,0)\Pi'(y,0)dy + \int_0^t \int_{\mathbb{R}^N} \Pi' \hat{L} G dyds$$

- This allows us to pull out Π' from the last term, using fact $\hat{L}G = \delta(x-y)\delta(t-s)$ and the sifting property.

$$\Pi'(x,t) = \int_{\mathbb{R}^N} G(x,t;y,0)\Pi'(y,0)dy + \int_0^t \int_{\mathbb{R}^N} G(x,t,y,s)\nabla_y \cdot (\left\langle v'(y,s)\Pi'(y,s)\right\rangle - v'(y,s)\Pi(y,s))dyds$$

- Finally, multiply by v' and take the ensemble mean, we arrive at an unclosed exact expression for the cross variance term:

$$\left\langle v'\Pi'\right\rangle(x,t) = \int_{\mathbb{R}^N} G(x,t;y,0)\left\langle v'(x,t)\Pi'(y,0)\right\rangle dy - \int_0^t \int_{\mathbb{R}^N} G(x,t;y,s)\nabla_y \cdot \left\langle v'(x,t)(v')^T(y,s)\Pi(y,s)\right\rangle dyds$$

- By taking ensemble we have wiped out some $\langle \Pi' \rangle$ terms, for any remaining Π' terms, the LED closure simply takes the approximation by the mean-field:

$$\Pi' \approx \langle \Pi' \rangle$$

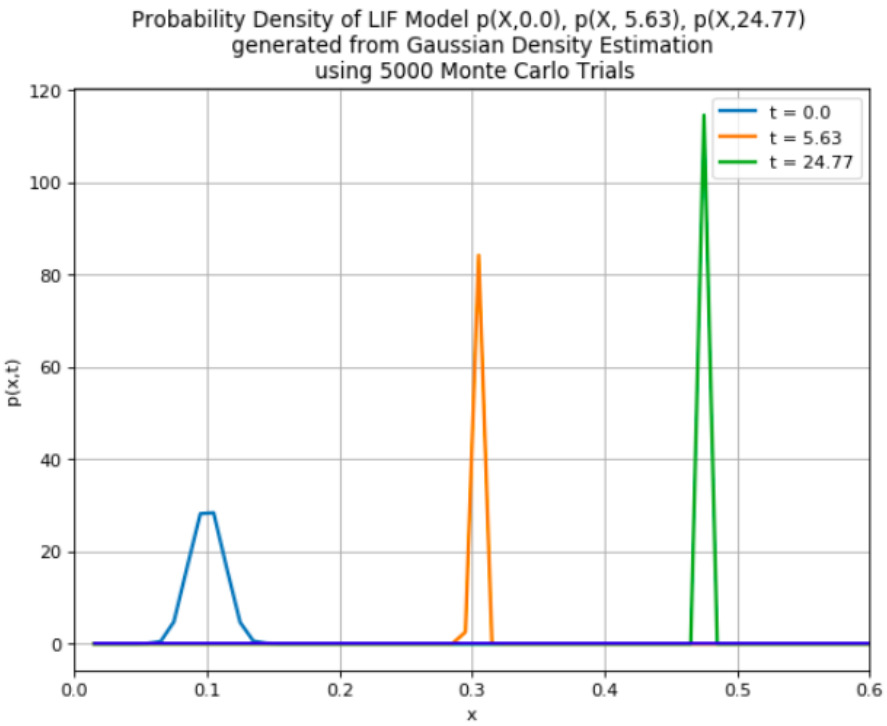
- We omit the detail here, applying *Liouville-Ostrogradski lemma*, by noting the properties of G and the flow map $\phi(s) = y$, we finally arrive at:

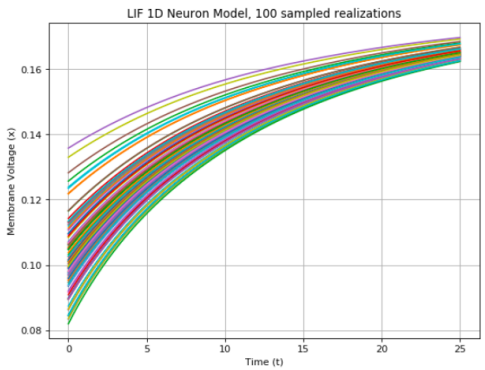
$$\left\langle v'\Pi' \right\rangle (x,t) \approx -\int_0^t \exp\left(-\int_s^t \nabla_\phi \cdot \langle v(\phi(r),r)\rangle \, dr\right) \nabla_\phi \cdot \left(\left\langle v'(x,t)(v')^T(\phi(s),s)\right\rangle p(\phi(s),s)\right) ds$$

- In particular, the exponential part is the Jacobian determinant of the reverse flow map. As the flow map is dependent on time, we have a time-ordered matrix exponential (very difficult to compute numerically)
- Modified LED closure arguments attempt to make the integro-differential PDE more computable by making more assumptions (skipped)
  - T.Maltba, P.A. Gremaud, D.M. Tartakovsky, Nonlocal PDF methods for Langevin equations with colored noise, Journal of Computational Physics
- The semi-localized method achieved best results in both computational time and accuracy (on test problems shown in the numerics section)
  - one way is to treat ignore the fluctuations of p within $(t-\tau, t)$
  - Has limitations

Data-Driven Equation Discovery

- The main purpose is to significantly reduce the number of MC simulations needed to converge to the true solution.
- The data-driven method using LASSO regression achieved 80% to 95% reduction of Monte-Carlo trials.





Setup of Regression

- Reformulate our general problem of form:

$$\frac{\partial p}{\partial t} + \nabla_X \cdot (\langle v \rangle p) + \mathcal{L}_X(p; \beta) = 0 \tag{6}$$

where $\mathcal{L}(\cdot)$ is an unknown linear differential operator which we seek to approximate from a dictionary of numerical derivative features.

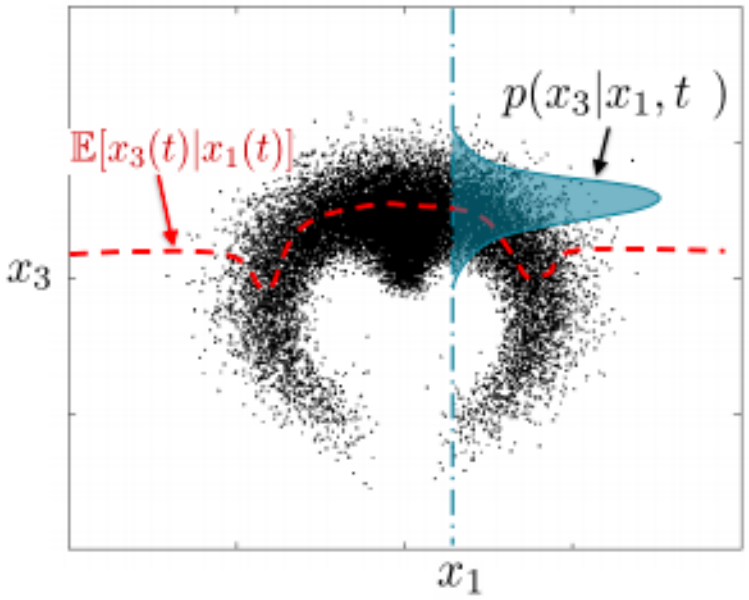


Figure: An example of using regression on Monte-Carlo data for term discovery

Overview of Regression Problem

- An example of such a dictionary [2] is:

$$\mathcal{L}_X = \beta(X,t) \cdot \left[1, \frac{\partial}{\partial X_1}, \cdots, \frac{\partial}{\partial X_N}, \frac{\partial^2}{\partial X_1^2}, \cdots \frac{\partial^2}{\partial X_N^2}, \frac{\partial^2}{\partial X_1 X_2}, \cdots\right]$$

- We assume the numerical/data-generated PDF $\hat{p}$ is defined on a uniform mesh
- The problem is physically-informed and can be reduced to a conservative form $\hat{\mathcal{L}}_X$; we define the residual to be:

$$\mathcal{J}(\beta) = \hat{p}_t + \nabla_X \cdot (\langle v \rangle \hat{p}) + \nabla_X \hat{\mathcal{L}}_X(\hat{p}; \beta)$$

Then the regression is:

$$\hat{\beta} = \underset{\beta}{\operatorname{argmin}} \{ \|\mathcal{J}(\beta)\|_2^2 + \lambda \|\beta\|_1^2 \}$$

where the L_1-norm penalty promotes sparsity in the solution.

Revisiting Kramers-Moyal Expansion

Theorem (Pawula)

Let:

$$\frac{\partial p(X,t)}{\partial t} = \sum_{i=1}^{\infty}(-\frac{\partial}{\partial X})^n D^{(n)}(X,t)p(X,t) = \mathcal{L}_{KM} \cdot p(X,t)$$

be the Kramers-Moyal expansion for $p(X,t)$ and $\mathcal{L}_{KM}(\cdot)$ is the Kramers-Moyal operator.
The Pawula theorem states that for a valid probability density p, the expansion either stops after the first order term or the order second term, or it must contain an infinite number of terms.

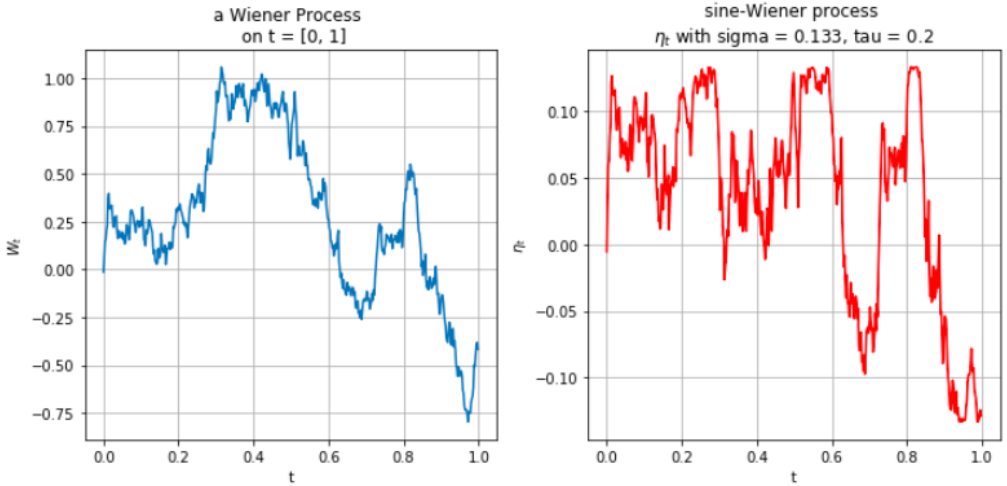
This greatly shrinks our search space.

Numerical Examples

- The noise term in our examples is *sine-Wiener* noise [5], given by the following expression:

$$\eta_t = \sigma \sin\left(\sqrt{\frac{2}{\tau}} B_t\right)$$

where σ, τ, the noise amplitude and correlation timescale, are positive. And B_t is the standard Brownian motion.



- The covariance has the formula (for $s \leq t$):

$$\langle \eta_t \eta_s \rangle = \frac{1}{2}\sigma^2 \exp\left(\frac{-(t-s)}{\tau}\right)\left[1 - \exp\left(\frac{-4s}{\tau}\right)\right]$$

We tested our method on problems with 1D and 2D spatial dimensions:

- (1D) Leaky Integrate and Fire Neuron Model

$$\dot{x_t} = \frac{\sigma^2}{2x_t} - \mu x_t + \eta_t$$

- (2D) Stochastic Fitz-Hugh Nagumo Neuron Model

$$\begin{pmatrix} \dot{x_1} \\ \dot{x_2} \end{pmatrix} = \begin{pmatrix} x_1 - \frac{x_1^3}{3} - x_2 + I \\ \epsilon(x_1 + \alpha - \beta \cdot x_2) + \eta_t \end{pmatrix}$$

Leaky Integrate and Fire Model

- A suitable closure approximation of the cross-variance term is:

$$\nabla_X \cdot \langle v'\Pi' \rangle \approx \hat{\beta}_1 p_X + \hat{\beta}_2 p_{XX}$$

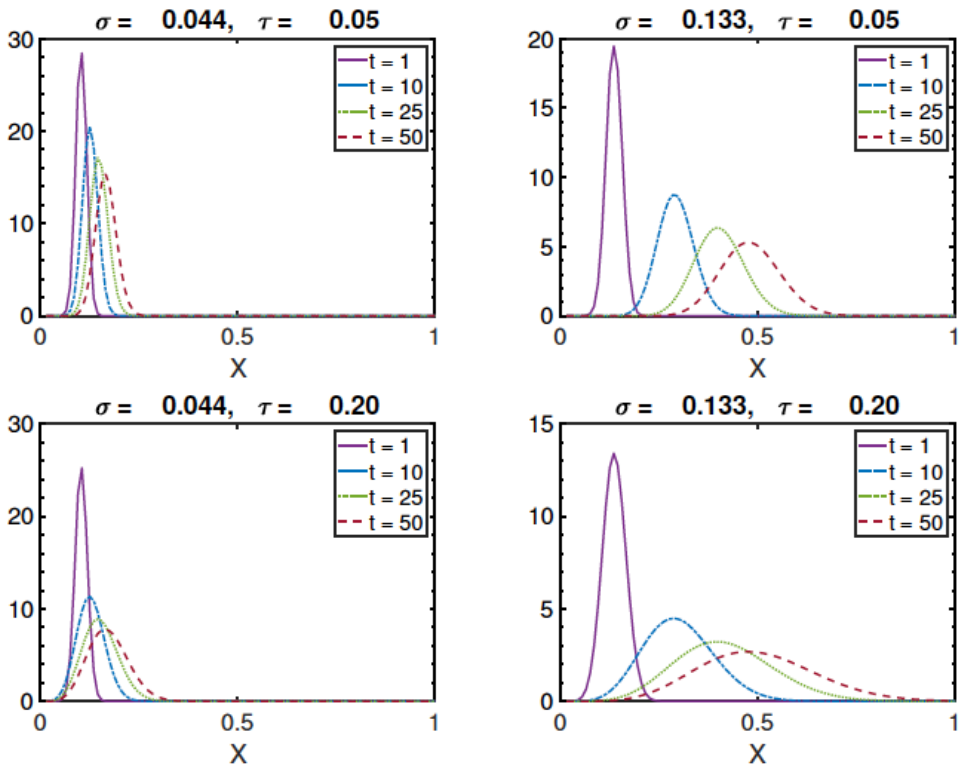


Figure: Evolution of the PDF p computed overtime

1D Numerical Results, Data Driven

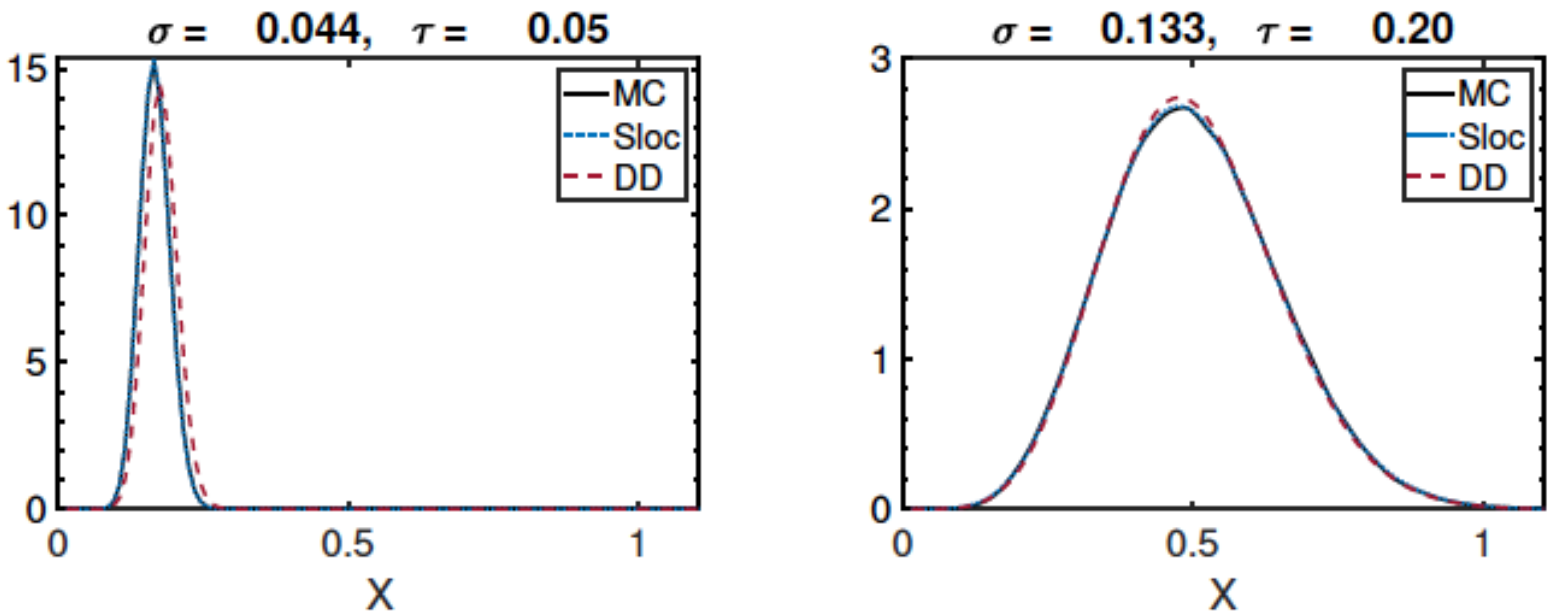


Figure: Comparison of p given by Data-Driven Closure and Analytical Closure, at final time $t = 25$

Stochastic Fitz-Hugh Nagumo Model

- The data-driven closure approximation of the cross-variance term used was:

$$\nabla_X \cdot \langle v'\Pi' \rangle \approx \hat{\beta}_1 p_{X_1} + \hat{\beta}_2 p_{X_2} + \hat{\beta}_3 p_{X_1 X_1} + \hat{\beta}_4 p_{X_1 X_2} + \hat{\beta}_5 p_{X_2 X_2}$$

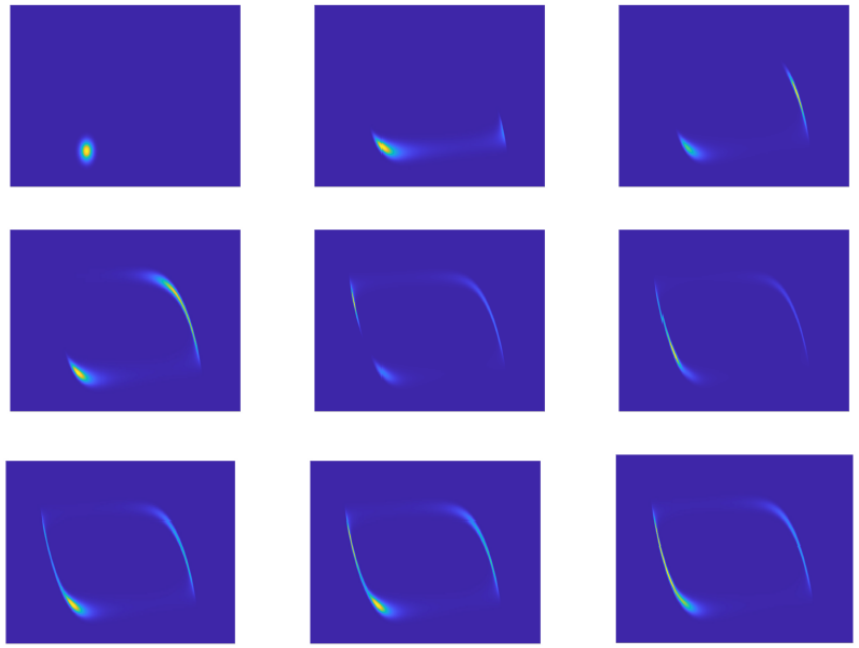


Figure: Evolution of the joint PDF at $t = 0, 5, 10, 15, 20, 30, 55, 60, 70$. X_1, X_2 are the horizontal and vertical axes, respectively

2D Numerical Results, continued

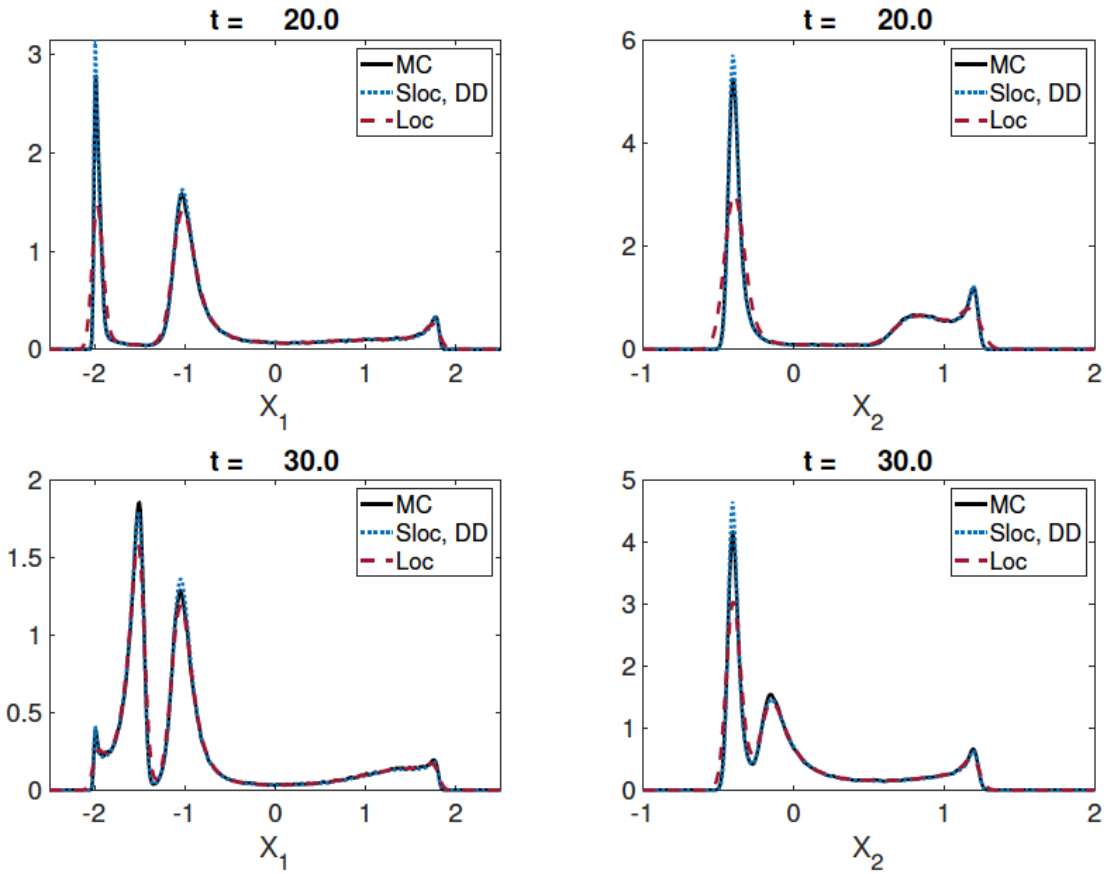


Figure: Comparison of marginal distributions computed computed using analytical approximation (Loc), and the Data Driven method, against Monte-Carlo estimation, with $\sigma = 0.05, \tau = 0.1$

Neural Network Solution

Reference: [Lagergren et al. 2020] *Biologically-informed neural networks guide mechanistic modeling from sparse experimental data*

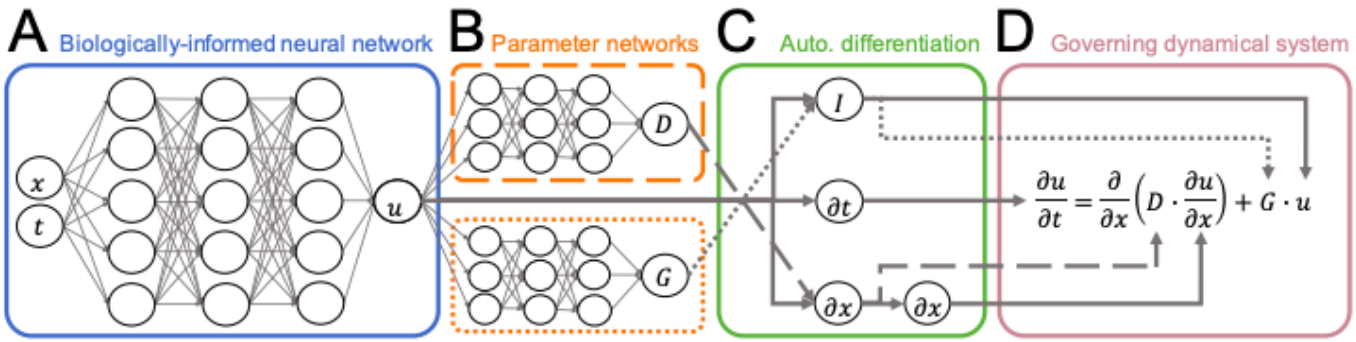


Fig 2. Biologically-informed neural networks for reaction-diffusion models.

Figure: flow chart of u_{MLP}

Conclusion and Work-in-Progress

We considered a data-driven model for discovery of nonlinear dynamics based on regression.

- Compare performance of neural network learning
- Consider reduced-order equations in high-dimensional setting

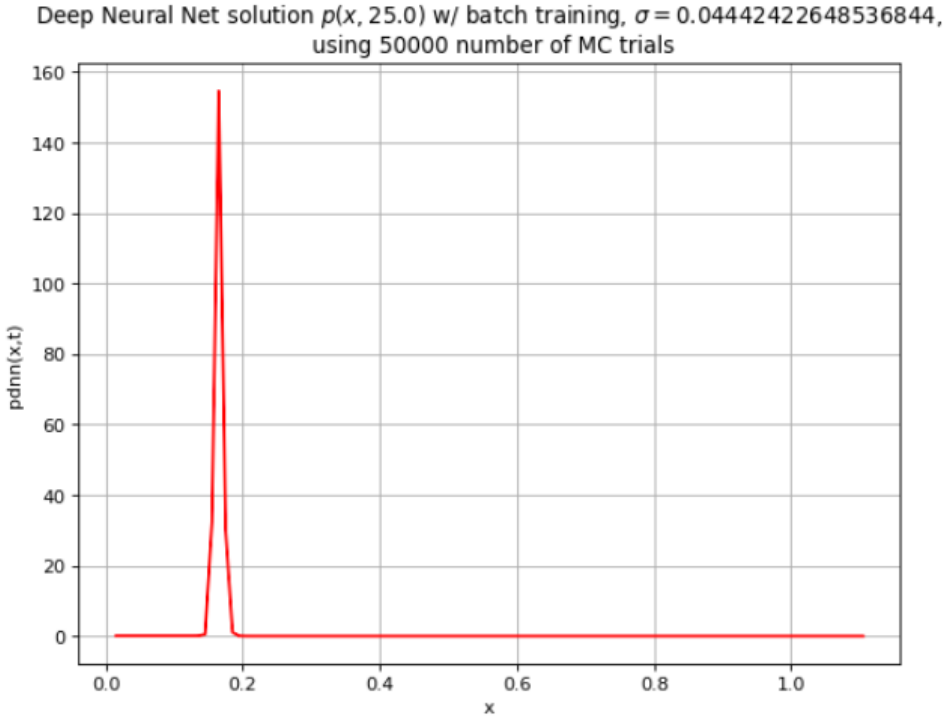


Figure: Deep Neural Net approximation of Monte-Carlo PDF using 80% of training data

Reference

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Questions

Thanks for listening!

Appendix A: Miscellaneous Facts

- Each realization from Monte-Carlo simulation was achieved using a 4-th order Runge-Kutta time-integration
- The conservative-advection equation is solved using *Clawpack*, a package of finite volume methods for linear and nonlinear hyperbolic systems of conservation laws, using a Lax-Wendroff scheme with MC limiter.
- The numerical schemes used for the 1D and 2D problems in time are respectively, Crank-Nicholson and Modified Craig-Sneyd ($\theta = \frac{1}{2}$) scheme